Name:_____

Displacement, Velocity & Acceleration

1. Q: A car moved 80 km to the South. What is its displacement?
A: 20 km South
B: 50 km East
C: 80 km South
D: 160 km North

2. Q: A car moved 60 km East and 90 km West. What is the distance?
A: 30 km
B: 60 km
C: 90 km
D: 150 km

3. Q: A car moved 60 km East and 90 km West. What is the displacement?
A: 30 km West
B: 60 km West
C: 30 km East
D: 150 km

4. Q: Average velocity can be calculated by dividing displacement over what?
A: time
B: distance
C: mass
D: density

5. Q: What is the average velocity of a car that moved 60km in 3 hours?
A: 10 km/h
B: 20 km/h
C: 30 km/h
D: 60 km/h

6. Q: What is the average velocity of a car that moved 40 km East and 80 km West in 2 hours?
A: 5 km/h
B: 10 km/h
C: 15 km/h
D: 20 km/h

7. Q: How far will a car travel in 25 min at 12 m/s?
A: 10 km
B: 14 km
C: 18 km
D: 24 km

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8. Q: How far will a car travel in 2 hours at 20 m/s?
A: 144 km
B: 158 km
C: 168 km
D: 234 km

9. Q: If car A is at 40 km/h and car B is at 10 km/h in the opposite direction, what is the velocity of the car A relative to the car B?
A: 10 km/h
B: 20 km/h
C: 40 km/h
D: 50 km/h

10. Q: If you are walking at constant velocity of 8 km/h and a car passed you by at the speed of 30 km/h from behind, what is the car's velocity from your viewpoint?
A: 22 km/h
B: 30 km/h
C: 38 km/h
D: 40 km/h

11. Q: If car A is at 70 km/h and car B is at 50 km/h in the same direction, what is the velocity of the car A relative to the car B?
A: 10 km/h
B: 20 km/h
C: 30 km/h
D: 40 km/h

12. Q: If a car moves 12 km North, 19 km East, and 12 km South, what is its displacement?
A: 12 km
B: 19 km
C: 31 km
D: 43 km

13. Q: Acceleration is the measure of the change in what?
A: density
B: motion
C: velocity
D: mass

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14. Q: Average acceleration is calculated by:
A: velocity change divided by the mass
B: mass change divided by elapsed time
C: velocity change divided by elapsed time
D: velocity change divided by gravity

15. Q: If a car accelerates from 3 m/s to 12 m/s in 3 seconds, what is the car's average acceleration?
A: 1 m/s^2
B: 2 m/s^2
C: 3 m/s^2
D: 4 m/s^2

16. Q: How long does it take to accelerate an object from rest to 15 m/s if the acceleration was 3 m/s^2?
A: 2 s
B: 4 s
C: 5 s
D: 15 s

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17. Q: You started to run at 10 km/h when you left your house and you arrived at school 30 minutes later. Assuming that your average acceleration was 30 km/h^2, how fast were you running when you arrived?
A: 15 km/h
B: 20 km/h
C: 25 km/h
D: 30 km/h

18. Q: Acceleration due to gravity is represented by what symbol?
A: v
B: g
C: t
D: s

19. Q: What is g?
A: 9.8 m/s^2
B: 10.7 m/s^2
C: 12.6 m/s^2
D: 98 m/s^2

20. Q: How long will it take for an apple falling from a 29.4m-tall tree to hit the ground?
A: 1.56 s
B: 2.04 s
C: 2.45 s
D: 3.72 s

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21. Q: How long does it take for a car to change its velocity from 10 m/s to 25 m/s if the acceleration is 5 m/s^2?
A: 2 s
B: 3 s
C: 4 s
D: 5 s

22. Q: If a car has a constant acceleration of 4 m/s2, starting from rest, how fast is it traveling after 5 seconds?
A: 20 m/s^2
B: 24 m/s^2
C: 30 m/s^2
D: 40 m/s^2

23. Q: If a car has a constant acceleration of 4 m/s2, starting from rest, how far has it travelled after 5 seconds?
A: 50 m
B: 60 m
C: 70 m
D: 80 m

24. Q: If a car has a constant acceleration of 4 m/s^2, starting from rest, how far has it travelled by the time it reaches the speed of 40 m/s?
A: 50 m
B: 100 m
C: 200 m
D: 400 m

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25. Q: A car is at velocity of 20 km/h. If the car travelled 120 km in 3 hours at constant acceleration, what is its final velocity?
A: 50 km/h
B: 60 km/h
C: 70 km/h
D: 80 km/h

26. Q: How long will it take for a falling object to reach 108 m/s if its initial velocity is 10 m/s?
A: 6 s
B: 8 s
C: 10 s
D: 12 s

27. Q: What is the final velocity of an apple if it falls from a 100m-tree?
A: 43.4 m/s
B: 44.3 m/s
C: 45.7 m/s
D: 46.4 m/s

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28. Q: What is the displacement of a car whose initial velocity is 5 m/s and then accelerated 2 m/s^2 for 10 seconds?
A: 150 m
B: 175 m
C: 200 m
D: 250 m

This one is a tricky one. You will need to do some working out below

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29. Q: What is the final velocity of a car that accelerated 10 m/s^2 from rest and travelled 180m?
A: 30 m/s
B: 40 m/s
C: 50 m/s
D: 60 m/s

30. Q: If a car accelerated from 5 m/s to 25 m/s in 10 seconds, how far will it travel?
A: 100 m
B: 150 m
C: 200 m
D: 250 m

Solutions

1. C: 80 km South
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2. D: 150 km
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3. A: 30 km West
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4. A: time
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5. B: 20 km/h
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6. D: 20 km/h
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7. C: 18 km
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8. A: 144 km
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9. D: 50 km/h
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10. A: 22 km/h
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11. B: 20 km/h
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12. B: 19 km
---
13. C: velocity
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14. C: velocity change divided by elapsed time
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15. C: 3 m/s2
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16. C: 5 s
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17. C: 25 km/h
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18. B: g
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19. A: 9.8 m/s2
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20. C: 2.45 s

21B: 3 s
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22. A: 20 m/s2
---
23. A: 50 m
---
24. C: 200 m
---
25. B: 60 km/h
---
26. C: 10 s
---
27. B: 44.3 m/s
---
28. A: 150 m
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29. D: 60 m/s
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30. B: 150 m
